Practical - 1

Practical Title: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

A. Theory

1. What is URL and parts of URL?

Answer:

A URL (Uniform Resource Locator) is a unique address used to locate and access resources on the internet.

- Identifies the location of a resource online.
- Specifies the protocol (e.g., HTTP or HTTPS).
- Includes the domain name or IP address.
- May contain a path to a specific resource.
- Used to access web pages, files, images, and other online resources.

PARTS OF URL

- Scheme: Defines the protocol used to access the resource (e.g., http, https, ftp).
- Subdomain: Specifies a subdivision of the main domain (e.g., www).
- Domain: Identifies the main website name (e.g., example).
- Top-Level Domain (TLD): Indicates the domain extension (e.g., .com, .org).
- Port Number: Specifies the server port (default: 80 for HTTP, 443 for HTTPS).
- Path: Points to the resource location on the server.
- Query String Separator: The ? symbol that starts query parameters.
- Query String / Parameters: Passes data as key-value pairs using &.
- Fragment: Refers to a specific section within the resource using

https://www.geeksforgeeks.org/computer-networks/components-of-a-url/

2. How does HTML file process in web browser?

Answer:

You Open the File in a Browser
When you double-click the file or open it through a web server, a browser like Chrome, Firefox, or Safari reads the code.The Browser Parses the HTML
The browser scans your code from top to bottom. It starts with the <! DOCTYPE html> declaration, then reads the file and then browser builds the page and display the content.

https://dev.to/ofodile/how-html-works-in-the-browser-25pa

3. How will page navigation flow managed among all HTML pages?

Answer:

Each page has links to the other pages, such as Home, Courses, Assignments, Contact Us, and Profile. When a user clicks a link, the href attribute opens the required HTML page.

4. How will GitHub commits be maintained after each practical?

Answer:

After each practical, you should maintain your GitHub commits by staging your completed files, writing a clear descriptive commit message, and pushing your local updates to the remote GitHub repository.

https://github.com/git-guides/git-commit

B. Evidence of Work (Screenshots)

 STUDENT HUB

Welcome to Your Learning Portal

Home | Courses | Assignments | Contact Us | Profile

| Welcome Student! |

Access all your academic information from one place

| 📖 Courses | | 📝 Assignments | | 📂 Materials |

View Courses Check Tasks View Notes

[Open] [Open] [Open]

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C. GitHub Repository Link:

https://github.com/Dhanraj2408/StudentHub

D. Challenges Faced:

- Deciding the proper layout and placement of elements on each page.
- Maintaining the same design and style across all 10 pages.
- Creating a simple and easy-to-use navigation flow.
- Managing different sections like Courses, Assignments, Notes, and Profile.
- Making the design attractive while keeping it simple and user-friendly

E. Learning Outcomes:

- Understood how different pages are connected through navigation and sitemap.
- Improved knowledge of creating wireframes and arranging UI elements properly.
- Learned to maintain consistency in fonts, colours, buttons, and page layouts.
- Got practical experience in designing a multi-page StudentHub portal.

Faculty Remarks: (Keep this blank for faculty)